

BIRLA INSTITUTE OF TECHNOLOGY & SCIENCE, PILANI –K. K. BIRLA GOA CAMPUS
INSTRUCTION DIVISION
FIRST SEMESTER 2026-2027
COURSE HANDOUT (PART II)

August 1, 2026

In addition to part-I (General Handout for all courses appended to the time-table), this portion gives further specific details regarding the course.

Course No. : EEE/INSTR F313
Course Title : Analog & Digital VLSI Design
Instructor-in-charge : PRAVIN MANE
Team of Instructors : Debasis Das (Office Room No. D204, debasisdas@goa.bits-pilani.ac.in),
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1 Scope and Objective of the Course:

The objective of this course is to provide to the students an introduction to the fundamentals and practical considerations pertaining to the design of integrated circuits. The scope encompasses both analog and digital integrated circuits. The importance of CAD tools in IC system design process is also acknowledged and stressed upon.

2 Text Books:

1. Behzad Razavi, "Design of Analog CMOS Integrated Circuits", TATA McGRAW Hill, 2001.*
2. Kang S.M. and Leblebici Y., "CMOS Digital Integrated Circuits: Analysis and Design", McGraw Hill International Editions 3rd Edition 2003.**
3. Neil H.E. Weste, Kamran Eshraghian, "Principles of CMOS VLSI Design", Addison-Wesley Publishing Company.***

3 Prime Reference Book:

1. Rabaey J. M., Chandrakasan Anantha, Nikolic Borivoje, "Digital Integrated Circuits : A Design Perspective", PHI Learning Private Limited, 2nd Edition.

4 Other Reference Books:

1. Pucknell D.A., Eshraghian K., "Basic VLSI design, systems and circuits", Third edition, Prentice Hall of India Pvt. Ltd.
2. Fabricius E.D., "Introduction to VLSI design", McGraw Hill international editions.
3. Gregorian R., Temes G.C., "Analog MOS integrated circuits for signal processing", Wiley interscience publication.
4. Sze S.M., "VLSI Technology", Second edition, McGraw Hill International Edition.
5. Martin. Ken, "Digital Integrated Circuit Design", Oxford University Press, Inc.
6. Johns. David A. and Martin K, "Analog Integrated Circuit Design", John Wily & Sons Inc. 2002.

5 Course Plan:

Lect. No.	Topic	Learning Objective	Book Reference
1	Operational Amplifier Performance parameters	Gain, small signal bandwidth, large signal behavior, output swing, linearity, noise, offset, supply rejection	9.1*
2	Single stage Op Amp	Analysis of single stage Op Amp	9.2.1*
3	Single stage Op Amp	Design procedure for single stage Op Amp	9.2.2*, 9.2.3*
4	Folded cascode Op Amp	Analysis & Design of folded cascode Op Amp	9.2.4*, 9.2.5*
5	Two stage Op Amp	Analysis & Design of two stage Op Amp	9.3*
6	Gain boosting	Basic idea and circuit implementation, frequency response	9.4*
7	Common Mode Feedback	basic concept, sensing techniques	9.7.1*, 9.7.2*
8	Common Mode Feedback techniques	Circuit implementations	9.7.3*
9	Other properties of Op Amp	Input range limitations, slew rate	9.8*, 9.9*
10	Other properties of Op Amp	power supply rejection, noise	9.11*, 9.12*
11	Stability in Op Amp	General considerations, Multi-pole systems	10.1*, 10.2*
12	Stability in Op Amp	Conditions for stability, concept of phase margin and gain margin	10.3*
13	Frequency compensation techniques	General concept, Miller compensation	10.4*
14	Frequency compensation techniques	Compensation in two stage Op Amp	10.5*
15	MOS inverter-Static characteristics	Static parameters of inverters	5.1**

16	MOS inverter-Static characteristics	Inverters with different loads	5.2**,5.3**
17	MOS inverter-Static characteristics	CMOS inverters general characteristics, analysis and design of CMOS inverter	5.4**
18	MOS inverter-Switching characteristics	Switching parameters of inverter	6.1**,6.2**
19	MOS inverter-Switching characteristics	Calculation of delay times of the inverter	6.3**
20	Combinational MOS logic gates	General concepts, static analysis	7.1**, 7.2**
21	Combinational MOS logic gates	CMOS logic circuits	7.3**
22	Combinational MOS logic gates	Complex logic gates, Euler's path for layout minimization	7.4**
23	Delay in Combinational circuits	RC delay model	4.3***
24	Delay in Combinational circuits	Linear Delay Model	4.4***
25	Delay in Combinational circuits	Logical Efforts of paths	4.5***
26	Power in combinational circuits	Static and dynamic power	5.2***, 5.3***
27	Combinational Circuit Design	Static CMOS design	9.2.1***
28	Combinational Circuit Design	Ratioed circuits, Cascode Voltage Switch Logic	9.2.2***,9.2.3***
29	Combinational Circuit Design	Dynamic Circuits	9.2.4***
30	Combinational Circuit Design	Pass transistor logic	9.2.5***
31	Sequential Circuit Design	Sequencing static circuits, different methods	10.1***, 10.2.1***
32	Sequential Circuit Design	Max and min delay constraints	10.2.2***, 10.2.3***
33	Sequential Circuit Design	Clock skew	10.2.5***
34	Sequential Circuit Design	Circuit design of latches and flip-flops	10.3***
35	Adder architectures	basic circuit elements	11.2.1***, 11.2.2***
36	Adder architectures	Ripple carry adder, Carry skip adder, carry lookahead adder	11.2.2***
37	Binary multiplication	Array multiplication, booth encoding	11.9***
38	Array subsystem	SRAM basic concepts, circuit implementation	12.2***

6 Evaluation Scheme:

EC No.	Components	Duration	Marks	Date & Time	Remarks
1	Quiz-I	30 Minutes	20	21.08.2026 7.00-7.30 PM	Open Book
2	Quiz-II	30 Minutes	20	17.09.2026 7.00-7.30 PM	Open Book
3	Midsemester Examination	90 Minutes	60	09.10.2026 4.00-5.30 PM	Closed Book
4	Quiz-III	30 Minutes	20	30.10.2026 7.00-7.30 PM	Open Book
5	Quiz-IV	30 Minutes	20	20.11.2026 7.00-7.30 PM	Open Book
6	Comprehensive Examination	3 Hours	80	08.12.2026 AN	Closed Book
Total			200		

Out of 4 regular quizzes, best 3 will be considered in final total.

7 Make-up Policy:

Application for Make-up will be considered only for Mid-Sem and Comprehensive Examination. An application in writing with relevant certificates attached (medical from Campus Medical center or SWD) needs to be submitted to the IC of the course at least a day before the scheduled exam. No make-up will be given for tutorials and quizzes (no exceptions).

8 Attendance Policy:

Since BITS, Pilani is a residential campus; students are expected to engage in all regular and special lectures/tutorials as announced by the instructors. Attendance and class participation of the student will be considered during final grading and will be based on sole discretion of the instructors.

9 Grading notice

All students registered in the course are expected to appear for all evaluation components. Per section 4.19 of the BITS, Pilani Academic regulations, NC may be given to students if they fail to provide a chance for the instructor to evaluate their progress in the class. Absence in any evaluation

components without the prior consent of the instructor or submitting blank or incomplete/incoherent answer books may present grounds for awarding NC in the course.

A student should obtain 10% or more of the total marks to clear the course.

10 Honor code and disciplinary action

All submissions by students in this class towards quizzes, tests, and tutorials will be considered their original and individual work. It will be assumed that the students have not resorted to unfair means during the evaluation. If malpractice is discovered, the strictest action will be initiated against the student.

11 FAQs

- (i). *Is lecture/tutorial attendance compulsory?* – Attendance will be taken per instructor discretion. 100% attendance for lectures and tutorials is expected.
- (ii). *Can I get a makeup for tutorials/quizzes?* – No makeup will be provided for these. They are intended to serve as techniques to assess course speed and relevance.
- (iii). *I thought I understood all the tutorials; however, I did not do well in the examination. Why?* - Tutorial sessions are used as an aid to help understand concepts that have been taught in lecture and provide an opportunity for students to revisit topics with the instructor. They should not be viewed as templates for the mid-sem or comprehensive examination.
- (iv). *I find the course difficult. How do I improve my performance?* – Students are encouraged to meet with instructors during their office hours for additional help.

12 Announcements

Course management and announcements will be handled through the BITS moodle webpage. However, since 100% class attendance is expected, there might be a delay between class announcements to the information appearing on the moodle webpage.

13 Chamber Consultation Hour:

To be announced in class by individual instructor.

Instruction Incharge
EEE/INSTR F313